Suraj K Suresh

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Education

University of California, Santa Barbara — California, USA
Masters in Computer Science; GPA: 3.71 — *2023 – present*

Amrita School Of Engineering — Kerala, India
Bachelor of Technology in Computer Science; GPA: 8.17 — *2018 – 2022*

- Graduated magna cum laude: Received undergraduate degree with highest distinction.

Experience

Graduate Research Assistant - Seclab — Oct 2023 – Present
University of California Santa Barbara — *California, USA*

- Developed Java and Linux kernel–specific pipelines for **DARPA's AIxCC** competition with Team Shellphish.
- Built automated tooling for patch validation and bypass analysis leveraging **LLMs and fuzzing**.
- Contributed to multiple research grants focused on **Linux kernel fuzzing and exploitation**.

Research Scholar — Aug 2022 – Mar 2023
University of California Santa Barbara — *California, USA*

- Worked on a fuzzing project that focused on finding **double fetch (TOCTOU) bugs in the Linux kernel**. Under the guidance of Prof. Giovanni Vigna and Prof. Christopher Kruegel.

Security Researcher Intern — Apr 2021 – Oct 2021
Cloudfuzz,Payatu — *Pune, India*

- Developed core functionality of the Cloudfuzz platform which involved writing test cases, harnesses and automating the deployment of the Cloudfuzz platform
- Worked on **Reverse Engineering** projects of embedded devices as part of Payatu

Student Scholar — Jun 2020 – Aug 2020
Google Summer Of Code 2020, ReactOS — *India*

- Worked on integrating **Syzkaller** with the **ReactOS codebase** involving sanitizers for efficient fuzzing.

Projects

Artiphishell (Github) | *Fuzzing, LLM, Binary analysis* — Nov 2023 – Jun 2025

- Contributed to development of a Java pipeline leveraging static analysis (semgrep, codeql) and sink-driven fuzzing using Jazzer.
- Built a Linux kernel pipeline integrating Syzkaller and Snapchange for directed and snapshot fuzzing.
- Created an automated analysis framework to test generated patches for bypasses and produce reports.

Binary Diffing for Automated Analysis | *Binary analysis, IDAPython* — Dec 2023 – Feb 2024

- Developed IDA-based tooling for large-scale binary diffing and analysis.
- Implemented automated flagging of binaries based on decompiled and disassembled code differences.
- Tooling incorporated into research on Android system and vendor update analysis (paper).

secREtary (Github) | *Pintool, Binary analysis, Reverse Engineering* — Jun 2020 – Sep 2020

- Implemented the logging module that traces function calls and prints out function statistics.
- Implemented a VManalyze module that dumps the instruction switch case table and other statistics.

ReactOS | *Fuzzing, Reverse Engineering* — Dec 2019 – Dec 2020

- Wrote initial grammar definitions for the ReactOS kernel and added code to ReactOS kernel to be compatible with fuzzer syscalls.
- Made a port of Syzkaller that fuzzes Windows Kernel using the NTDLL layer and wrote detailed blog posts explaining about progress and challenges faced.

Technical Achievements

Defcon-33 CTF Finalists & Defcon-32 AIxCC Finalists — *Shellphish*
Finisher of Flare-on 8 Challenge — *Individual*
Runner-up in ISITDTU CTF Finals — *teambi0s*
Finalists in CSAW CTF Nationals — *teambi0s*
5th Rank Decompetition CTF — *teambi0s*